

NOAKHALI SCIENCE AND TECHNOLOGY UNIVERSITY
INSTITUTE OF INFORMATION TECHNOLOGY (IIT)
Research Progress Summary

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Session: 2020–2021

1 Thesis Title

Enhanced Urban Mobility through Deep Learning Based Real-Time Passenger Head Counting and Occupancy Estimation

2 Research Question

- Can head detection be used to count passengers accurately in bus interiors?
- Which detection model performs best for small head detection in real bus scenes?
- Can tracking improve counting stability across frames and reduce double counting?

3 Methodology

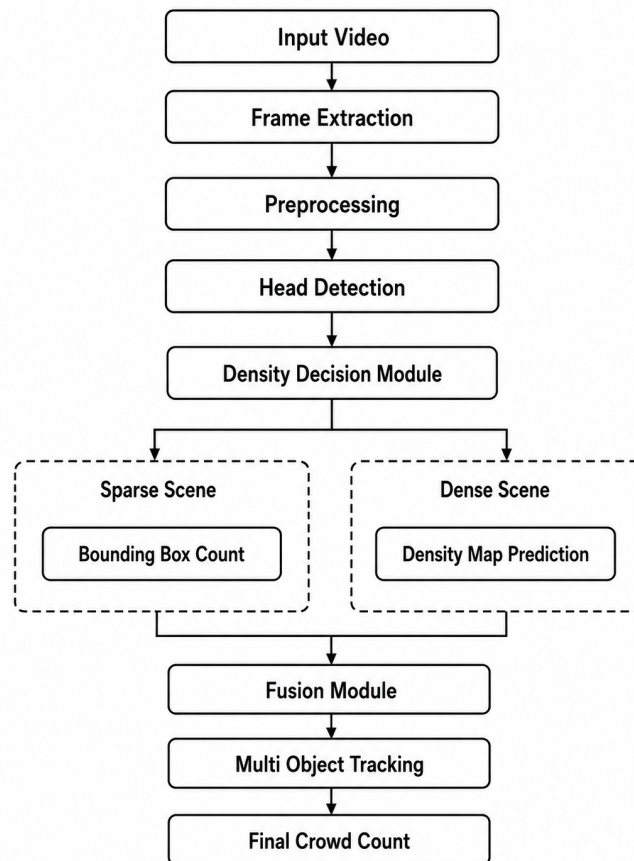


Figure 1: System Architecture

4 Meeting Progress

Date	Topic	Discussion Details
21 April	Literature Review	Reviewed related works and identified research gaps.
27 April	Review Feedback	Revised the literature review based on supervisor feedback.
29 April	Methodology	Outlined the research implementation process.
17 June	Enhancement	Discussed techniques to improve performance.
20 June	Pipeline Evaluation	Received feedback on the implemented pipeline.

Table 1: Summary of Supervisor Meetings and Discussions